

The significance of protein sequence similarities

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Abstract

A general method of assessing the significance of scored best local alignments, particularly suited to protein sequence comparisons, is described. The method establishes the parameters describing the distribution of the best results from any search program, provided that the set is sufficiently large and the majority of the alignments arise from unrelated sequences. The expected frequency of occurrence of any score can then be calculated, together with the number of standard deviations above expectation. These provide sensible measures of significance without additional search operations. However the biological significance of any alignment or set of alignments does not solely depend on the improbability of the alignment, but on all relevant factors known to the biologist.

Introduction

The importance of detecting protein sequence similarities that convey biological insight to the molecular biologist can hardly be exaggerated. The international databases are now substantial in size. The flow of sequence information has accelerated rapidly in the last few years and will accelerate again with the advent of automatic sequencing devices. As many comparison methods report a set of best results for similarity, it becomes an important issue whether any particular alignment has significance. We describe here a simple and powerful method of analysing these results which provides quantitative values for the significance of alignments in cases where large numbers of comparisons have been reported between a query sequence and some reference set of sequences. The method can be applied whatever method is used to locate and score similar (sub)sequences. We generate the sets of best alignments for this study with an implementation of the 'Best Local Similarity' algorithm of Smith and Waterman (1981) on the I.C.L. 64 × 64 Distributed Array Processor (DAP; Flanders *et al.*, 1978), as previously described (Coulson *et al.*, 1987).

The basis of the method is a description of the form of the frequency distribution of alignment scores between 'unrelated' proteins. It is not practicable to predict this form analytically, because of the variable length of alignments and the fact that

they may contain insertions or deletions. In addition, the form of the distribution may be different for each query sequence and scoring scheme (and database). The significance of an alignment is most simply expressed as the probability that a given score would be reached in the comparison of the query sequence with a collection of unrelated proteins the size of the database being searched.

We illustrate the method with examples found using proglucagon as a query sequence and a range of PAM tables (Dayhoff *et al.*, 1978). These tables are effective tools for the location of the alignments with the greatest probability of being significant. The tables can be generated for different evolutionary distances (the number of PAMs or accepted point mutations found in an alignment of 100 residues) by repeatedly applying the primitive matrix of change probabilities defined for 1 PAM, to an initial population of amino acids. While this model for evolutionary changes may not apply exactly to any real case, the range from 20–250 PAMs provides a variety of sets of useful similarity scores. Other tables, e.g. based on physical or chemical parameters, can be used, provided they are expressed suitably for the local homology algorithm with a scale that provides a zero or negative expected score for random alignments.

It must be noted, however, that the significance of alignments lies in the information they convey to the biologist and not in their improbability. Alignments conveying information about probable features related to structure or function may be important, even though their scores are not outstanding and there is therefore a need to make available quite extensive lists of alignments for perusal and detailed analysis.

Algorithm

We are essentially trying to answer the question: What is the probability that a similarity as strong as that observed in some specific case will occur between unrelated proteins? Since in any large and non-specialized database, there is a wide variety of protein types, a single family of proteins can only provide a limited number of alignments. If the number of results considerably exceeds the number of alignments possible from a set of related proteins, we can use the distribution of the bulk of the results to assess the significance of the high-scoring alignments, since the majority of the lower scoring alignments must be with proteins we would regard as unrelated. We want to predict the behaviour of the extreme high-scoring end of the

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distribution, using the observed upper end results, without including results which may have arisen from related proteins. For the NBRF database, no family of proteins is represented much more than 100 times, and we can assume that the lower 97% of the reported 4096 results are alignments with unrelated proteins and can be used to derive the behaviour of the remainder of the set.

We demonstrate here that the upper end of the distribution, when only unrelated sequences are involved, should decay exponentially. Suppose we have an alignment, represented by a path through the match matrix, which has reached a score S . The alignment could be improved if, from any point reached with penalties X , it is possible, by permissible moves to find a continuation which can raise the score by more than X . The greater the loss of score, X , the less likely a net improvement becomes; the probability of improving an alignment is therefore determined largely by the chance of an improvement starting relatively close to the current end. This chance does not depend on the current value of S . If the probability of improvement is independent of S then the population of alignments scoring more than S will be a constant fraction of the number scoring S . That is, we have described a process which predicts exponentially declining populations of alignments as the score increases.

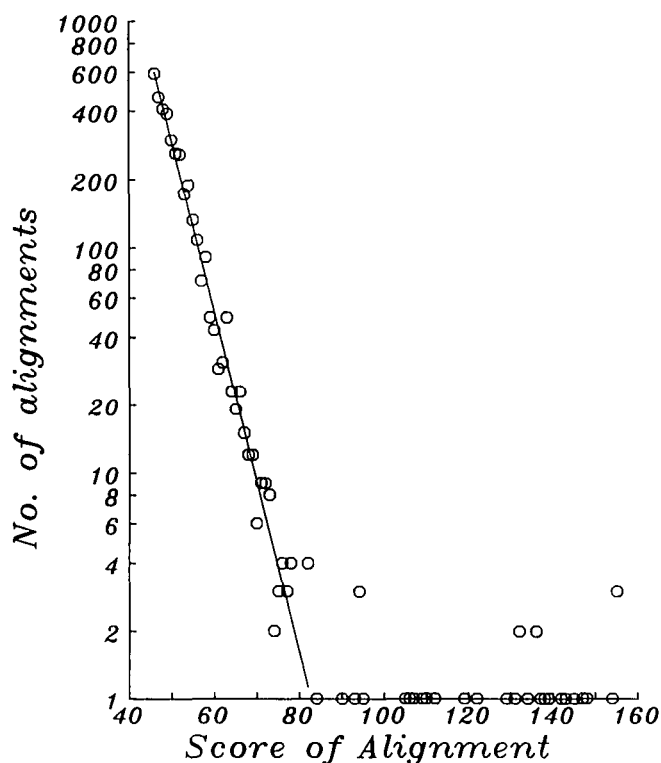


Fig. 1. Plot of the number of reported alignments achieving a given score (logarithmic scale) against the score value, for a search of the protein database with bovine proglucagon as the query sequence, using the 100 PAM similarity table. The straight line is fitted to the lowest 97% of the reported part of the distribution. The top 49 results represent unequivocally related sequences.

This expectation has been validated experimentally; Figure 1 shows a typical example. To assess the significance of the high-scoring alignments, we fit the best straight line to the logs of the numbers of observed alignments, against the score. Only classes containing the lower 97% of the alignments, arising from unrelated proteins, are used. The parameters obtained allow us to predict the expected frequency with which any specific score class should be observed. A search of the database with a query sequence derived by randomly shuffling a real protein sequence should generate a distribution of results which conforms throughout to the linear model. Figure 2 shows that this expectation is also borne out.

The expected frequency is the expectation that an alignment with a specified score occurs by chance in comparisons with a reference set of sequences as large as the database used, containing only unrelated sequences. This measure of significance is convenient and readily appreciated; it has the disadvantage that the expected frequency depends on the size of the database used. The larger this is, the more comparisons are made with unrelated proteins and the higher will be the scores of alignments reached by a chance similarity. This problem could be overcome by 'normalizing' the results to a standard-sized database (e.g. 1 000 000 residues), but a sounder alternative is to express the frequency of the score in terms of standard deviations from the fitted straight line; as the fits of the experimental

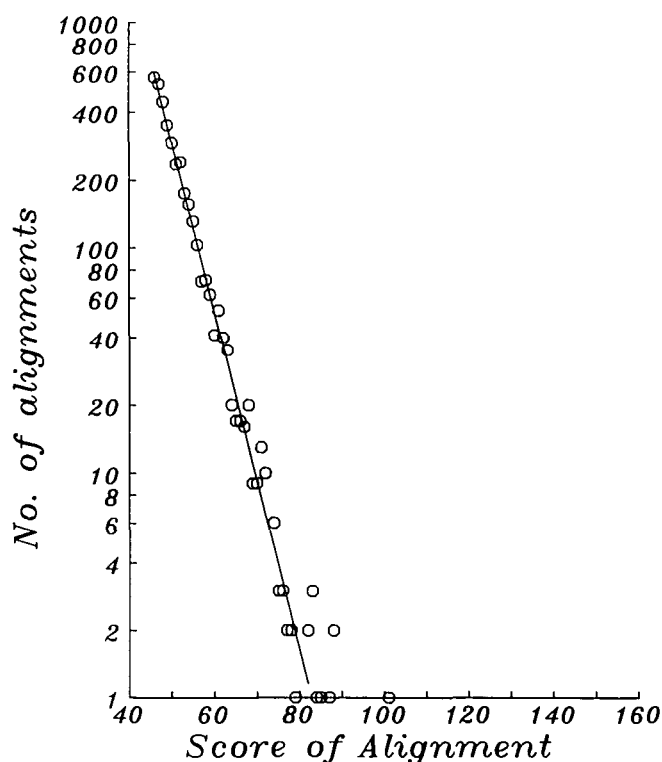


Fig. 2. As Figure 1, except that the query sequence was randomly shuffled before the search was performed. The parameters of the fitted straight line are almost identical to those in Figure 1.

points to a straight line are good, the standard deviations (s.d.) are small.

When the expected frequency falls below 1, the alignments merit attention, though it may not be possible to interpret them positively. We use a value of 0.01 to indicate alignments which need detailed examination. Many cases of significant alignments have now been demonstrated, with expectations that are orders of magnitude < 1 , and up to 60 s.d.s. above expectation (e.g., Robinson *et al.*, 1987).

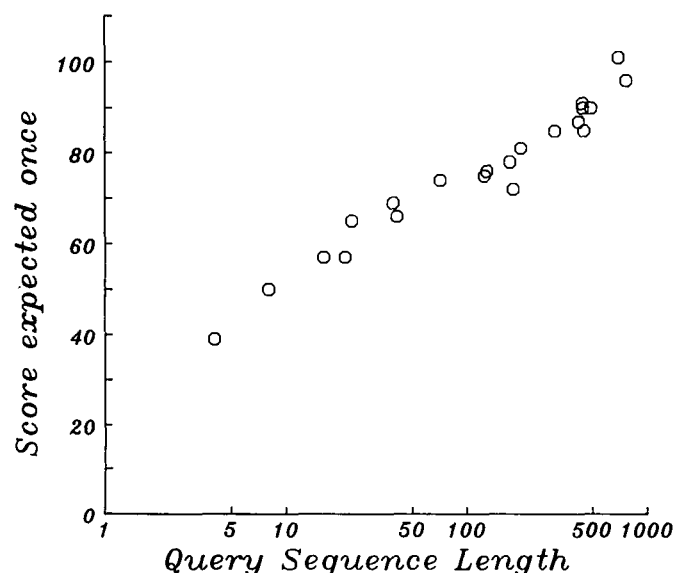


Fig. 3. Plot of the score expected once against the query sequence length (logarithmic scale) for 21 typical searches, using real proteins as query sequences, and searching the protein database with the 100 PAM similarity table.

Implementation

The alignment algorithms are particularly well-suited to parallel architecture computers (Lyall *et al.*, 1986) and details of the best local homology implementation on the I.C.L. DAP have been published (Coulson *et al.*, 1987). The program can exhaustively compare a 'query' sequence with every member of a protein sequence database and find all the locally-similar subsequence pairs, taking account of the possibility of any insertion/deletion at any position in either sequence.

The DAP returns ordered details of the alignments with the highest scores. More than one alignment can be reported from each comparison between a pair of sequences. The number of locally best alignments is huge and the top 4096 represent the tip of this much larger distribution. The results include only complete sets of results at each score and all scoring classes reported can therefore be used in the subsequent analysis.

We have made use of the similarity tables described by Dayhoff (1978), over the range 10–250 PAM's (accepted point mutations per 100 residues aligned). The indel penalty has been set to maintain the triangle inequality and to ensure that the proportion of gaps introduced into 'good' alignments is not excessive. The program requires about 1.8 s of DAP c.p.u. time per residue of the query sequence to complete a search of Release 11 (1 066 790 amino acid residues) of the NBRF database (George *et al.*, 1986).

The score of the alignment expected once, in the search with the NBRF Version 11 database and the 100 PAM similarity scoring table varies in proportion to the logarithm of the query length (Figure 3). An indication of what scores are significant could be obtained from the length alone, using this relationship.

20 PAM Table Hits = 19 Mismatches = 8 Indels = 0 Expected No. = 0.565402×10^{-5}				
** .***** * ** . * . *				
52	RHSQGTFTSDYSKYLSRRRAQDFVQWL	78		
77	RHAEGTFTSDVSSYLEGQAAKEFIAWL	103		
120 PAM Table Hits = 42 Mismatches = 15 Indels = 4 Expected No. = 0.252531×10^{-6}				
... ..*				
48	NED KRHSQGTFTSDYSKYLSRRRAQDFVQWLMNTK RNK NNIKRHDEFER HAEGTFT	104		
72	HDEFERHAEGTFTSDVSSYLEGQAAKEFIAWLVKGRGRDFPEEVNIVEELRRRHADGSFS	132		
160 PAM Table Hits = 58 Mismatches = 22 Indels = 4 Expected No. = 0.198303×10^{-11}				
... ..*				
48	NED KRHSQGTFTSDYSKYLSRRRAQDFVQWLMNTK RNK NNIKRHDEFER HAEGTFTSDVSSYLEGQAAKEFIAWLVKGR	127		
72	HDEFERHAEGTFTSDVSSYLEGQAAKEFIAWLVKGRGRDFPEEVNIVEELRRRHADGSFSDEMNYVLDLATRDFINWLLQTK	155		
250 PAM Table Hits = 59 Mismatches = 21 Indels = 3 Expected No. = 0.479972×10^{-6}				
. * ..***** * * . * . * . *				
48	NEDKRHSQGTFTSDYSKYLSRRRAQDFVQWLMNTK RNK NNIKRHDEFER HAEGTFTSDVSSYLEGQAAKEFIAWLVKGR	127		
73	DEFERHAEGTFTSDVSSYLEGQAAKEFIAWLVKGRGRDFPEEVNIVEELRRRHADGSFSDEMNYVLDLATRDFINWLLQTK	155		

Fig. 4. Alignments and expected frequencies reported for the first internal repeat of the bovine proglucagon sequence, using similarity tables from 20–250 PAM's. * indicates a matched pair of residues, . a pair with a positive score in the current table. A gap in the sequence indicates a postulated indel. Searches were also performed at 40, 60, 80, 100, 140, 180 and 200 PAM's, with similar results to those shown.

Discussion

We illustrate several features of this method with the protein proglucagon, without its signal peptide. This protein belongs to a large group of related proteins in the NBRF Protein database. The results of the DAP search show that the vast majority of the leading alignments were with proteins in the same family and that the internally repeated sequence segments gave additional alignments in the top category. The results with different PAM tables illustrate why the complexity of searching is often underestimated. The alignment between the gastrointestinal peptide from pig and the glucagon region in proglucagon is reported virtually unchanged between 20 PAMs to 250 PAMs. The expected frequency ('EF') of this alignment is least at 60 PAMs, at 2.6×10^{-6} ; by 250 PAMs the expected frequency has risen to 1.3, giving no indication of the importance of the homology indicated. A similar situation arises with secretin, another active peptide. The most significant result, also at 60 PAMs, had an expected frequency of 1.1×10^{-3} . This alignment is also reported from 40–200 PAMs (at 20 PAMs it is truncated somewhat). Above 60 PAMs the expected frequency rises; by 160 PAMs it is 0.18, by 200 PAMs 1.9, and by 250 PAMs it has risen so much that it is not reported among the top 50 alignments, which include many new alignments between proglucagon and probably unrelated proteins. Significant alignments do not all behave the same way; the alignment between proglucagon and the human vasoactive intestinal peptide is most significant at 140 PAMs, with an EF of 5×10^{-8} .

Another phenomenon encountered with the Best Local Homology algorithm of Smith and Waterman is the sensitivity of paths to the scoring scheme used. As the scoring scheme is changed, alignments may lengthen or shorten, and in extreme cases may fragment into distinct parts, with lower scores and less chance of being reported. This is seen in the behaviour of the first alignment between two repeated regions in proglucagon; at low PAMs an alignment of 27 residues is found, with EF about 3×10^{-6} ; by 100 PAMs an alignment of 42 residues is reported and at 160 PAMs the alignment jumps to 80 residues length, and EF 1.9×10^{-12} (Figure 4).

It is expected theoretically that the most sensitive scoring scheme in any particular case will be provided by the PAM table appropriate to the overall evolutionary distance in the alignment being sought, and these examples show this to be the case. Since the evolutionary distance is not generally known in advance, searches must be repeated with a range of PAM tables if significant similarities are not to be missed.

The ability to carry out database searches for similarities to known proteins, using the exhaustive matching algorithm for finding the best local homology, has proved a fruitful method of extending or guiding molecular biological research (Dorin *et al.*, 1987; Robinson *et al.*, 1987). We have described a method of assessment of the scores of alignments based only

on the results of the initial search operation. An alternative is to assess significance by reference to a further set of alignments produced by unrelated sequences—for example, scrambled versions of the test sequence. As these sequences give the same results in our analysis, they add little information to that already obtained.

The final test lies in the interpretation of the results by the biologist and it is clear that score alone does not determine the biological importance of an alignment. Thus, the cystic fibrosis antigen finds many alignments which are not individually significant, but which are all related to calcium-binding sites in proteins (Dorin *et al.*, 1987). In the case of proglucagon, many of the minor alignments confirm the presence of repeated sub-sequences in the whole family of glucagon precursors.

In practical terms, weak alignments may be better detected if the query sequence is presented in a series of small segments, perhaps 100 or less residues long, so reducing the background of alignments arising from regions not included in the specific region under study. Extremely short sequences cannot be expected to produce alignments significant by their score alone; other factors (e.g. the nature of the proteins in which matching sequences are found) must be used in interpreting the output.

We reiterate the general nature of this method; a complete database search is not required, merely the ability to accumulate a large number of scores from alignments which represent the top of the distribution of such alignments. We have demonstrated, for instance, that the method works effectively with a much smaller search (for example, with 20 000 amino acid residues in the reference sequence set), and the parameters still fit well to the model of exponential decay. Pattern searches (see Coulson *et al.*, 1987) produce similar parameters as searches with the same number of identified amino acid residues. Variable pattern spacings, with degenerate matching possibilities or with variable spacings, produce distribution with lower decay constants; i.e. a higher relative score must be achieved to be significant than with less flexible searches.

The parameters describing the upper portion of the distribution of results can also be applied to subsequent individual comparisons, since notionally each additional sequence could have been added to the original data searched without affecting the value of the parameters derived from the initial analysis.

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